Lecture Scribe Generation

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generate a lecture scribe suitable for exam preparation. white writing this, reason step by step through definitions, assumptions,, proofs, derivations and example solutions exactly as seen in the pdf. The instructions must follow exactly the logical flow used in the lecture material. Make every explicit so that we can clearly follow these during revision. Explain all the definitions and assumptions explicitly. Give step-by-step proofs, derivations and example solutions exactly as done in the ppt provided. Do not add new explanations, interpretations, examples, shortcuts and alternative methods. Maintain a formal, academic tone suitable for exam revision. Do not introduce any material that is not present in the given lecture ppts. Do not provide multiple explanations. Also do not summarize by skipping reasoning steps and also do not include personal comments or external knnowlegde. scan and see all the 18 pages of the pdf file and read everything and include all the topics from these slides, and give the content from the all these 18 pages, ie the entire pdf should be done at once with the format to be followed which is given above also dont sumaarize any points, keep them as given in the ppt and also make sure to include the contnt from the entire file read each and every line from the file from each slide slowly and then make the thing. read twice but make sure to include everything

and include the formulas as it is given in pdf nd no external knowledge but make sure to cover the entire file and give the entire scribe in one go and make sure not to group a bunch of slides and then summarize them together dont do that

Below is the **complete lecture scribe** constructed strictly from all 18 slides of the uploaded file .
The content follows the **exact logical progression of the lecture**, includes **all definitions, assumptions, formulas, derivations, and steps**, and does **not introduce any external material, interpretations, or shortcuts**.

Lecture 23: Predicting the Impossible – Tail Bounds and System Reliability

(Coding Session – Tails and Bounds)

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1. Introduction (Slide 1)

Title:
Predicting the Impossible: Tail Bounds and System Reliability

Context:

- Case Study: Amazon Black Friday Sale 2026
- Focus: Probability bounds, uncertainty, and system reliability

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2. Amazon Black Friday Sale 2026 (Slide 2)

Timeline (as given):

- Expected start: Friday, November 27, 2026
- Early deals: November 20–21, 2026
- Main sale period: November 25–27, 2026
- Cyber Monday: November 30, 2026

Target Matrix (as shown):

- Electronics & Tech → Discount ceiling: 75%
- Amazon Devices → Discount ceiling: 50%
- Fashion & Apparel → Discount ceiling: 60%
- Home & Kitchen → Discount ceiling: 60%

Additional factors:

- Amazon Prime advantages:
  - Early access to inventory
  - Exclusive cashback offers
- Partner bank leverage:
  - 10% instant discount
  - 5% multiplier cashback

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3. Connecting with Uncertainty and Computing (Slide 3)

From the diagram:

System Modeling

- Multiple servers: $S_1, S_2, S_3, S_4, \ldots$
- Each server processes jobs

Condition of interest:

- Highlighted server has **≥ k jobs**

Poisson Arrivals (Datacenter)

- Job arrivals modeled as a Poisson process
- Time window: 1 hour

Event of interest:

- Many arrivals within one hour
- Condition:

$$\text{Number of arrivals} \geq k$$

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4. Tail Probability Objective (Slide 4)

From the slide:

Goal:

$$P(X \geq k)$$

Interpretation:

- Probability that system load exceeds threshold

Context:

- Used to predict:
  - Server crashes
  - System overload

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5. Mean vs Tail Behavior (Slide 5)

Statement (as given):

- The mean is safe
- The tail is catastrophic

Simulation Details:

- Server traffic: 1000 minutes
- Distribution: Poisson with $\lambda = 50$

Threshold:

$$k = 60$$

Observation:

- Crash occurs when:

$$X \geq 60$$

Empirical crash rate:

- 87 crashes / 1000 minutes
- Crash rate = 8.7%

Required computation:

$$P(X \geq 60)$$

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6. Hierarchy of Bounds (Slide 6)

Statement:
Better inputs yield tighter bounds

Level 1: Markov's Inequality

- Input required: Mean μ

Level 2: Chebyshev's Inequality

- Input required: Variance σ^2

Level 3: Chernoff Bound

- Input required: Full distribution (MGF)

Insight:

- Exact probability may not be computable
- Upper bounds are used
- More information ⇒ less over-estimation

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7. Markov's Inequality (Slide 7)

Definition:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Interpretation:

- Requires only mean

Decay Behavior:

- Linear decay:

$$\frac{1}{k}$$

Observation from graph:

- Over-estimates significantly
- Labeled as:
  - "Wasted over-estimate"
  - "Decays linearly"

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8. Chebyshev's Inequality (Slide 8)

Definition:

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Equivalent form (as shown):

$$P(X \geq a) \leq \frac{\sigma^2}{(a-\mu)^2}$$

Decay Behavior:

- Polynomial decay:

$$\frac{1}{k^2}$$

Observation:

- Tighter than Markov
- Still over-estimates

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9. Chernoff Bound (Slide 9)

Definition:

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Where:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Alternative form (as shown):

$$P(X \geq a) \leq e^{-at + \ln M_X(t)}$$

Decay Behavior:

- Exponential decay:

$$e^{-k}$$

Observation:

- Tightest bound
- Closely matches exact probability

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10. Comparison of Decay (Slide 10)

Markov:

- Linear decay:

$$\frac{1}{k}$$

Chebyshev:

- Polynomial decay:

$$\frac{1}{k^2}$$

Chernoff:

- Exponential decay:

$$e^{-k}$$

Conclusion (as stated):

- Chernoff closely maps reality
- Markov remains pessimistic

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11. Case Study: SLA Breach (Slide 11)

Parameters:

- Distribution: Poisson
- Mean:

$$\lambda = 500$$

Threshold:

$$k = 600$$

Objective:

$$P(X \geq 600)$$

Task:

- Predict probability
- Size backup capacity

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12. Cost of Loose Bound (Slide 12)

Results (as shown):

- Exact risk: ~1.4%
- Chernoff: ~8.8%
- Chebyshev: ~20%
- Markov: ~90.9%

Interpretation:

- Markov predicts extremely high failure probability
- Leads to unnecessary provisioning

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13. Financial Impact (Slide 13)

Waste Multiplier:

$$\text{Waste Multiplier} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

At $\lambda = 500$:

- Chernoff: ~10.7× waste
- Chebyshev: ~6,422× waste
- Markov: ~107,039× waste

Conclusion:

- Loose bounds ⇒ massive cost

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14. Over-estimation Interpretation (Slide 14)

Statement:

Mathematical over-estimation = financial over-provisioning

Explanation:

- Over-estimated probability ⇒ more backup servers
- Leads to unused capacity

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15. Different Distributions (Slide 15)

Cases:

- Normal → stable jobs
- Poisson → independent requests
- Binomial → fixed users
- Gamma → bursty traffic

Observation:

- Chernoff works well for:
  - Poisson
  - Binomial

Limitation:

- For heavy-tailed (Gamma):
  - Over-estimation occurs
  - Alternative methods needed

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16. Decision Matrix (Slide 16)

Method	Requires	Decay	Tightness	Use Case
Markov	Mean μ	$1/k$	Loose	Rough check
Chebyshev	μ, σ^2	$1/k^2$	Moderate	Unknown distribution
Chernoff	Full MGF	e^{-k}	Tight	Precise sizing

17. Laplace Distribution Task (Slide 17)

Given:

$$X \sim \text{Laplace}(\mu, b)$$

Properties:

- Sharp central peak
- Heavy tails

Objective:

- Derive Chernoff bound

Threshold:

$$a = \mu + 3\sigma$$

Relation:

$$\sigma = b\sqrt{2}$$

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18. Derivation Blueprint (Slide 18)

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Chernoff Inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimize

- Differentiate w.r.t. t
- Set derivative = 0
- Solve for optimal t

Step 4: Substitute

- Plug optimal t into bound

End of Lecture Scribe

This scribe includes:

- All definitions
- All formulas exactly as given
- All steps and derivations in order
- All case study parameters and results

If you want next: I can convert this into **exam-ready LaTeX**, **handwritten-style notes**, or **expected questions + answers from this lecture**.